

Alice Lourenco

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EDUCATION

The University of North Carolina at Chapel Hill

Chapel Hill, NC

B.S. in Computer Science, Minor in Education

Aug 2025 – May 2029

Relevant courses: Algorithms, Data Structures, Object-Oriented Programming, System Analysis and Design

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, CUDA C++, Groovy, HTML/CSS, R

Frameworks: Spring Boot, FastAPI, React Native, Expo, Ruby on Rails

AI/ML: Claude API, RAG, sentence-transformers, LightGBM, pandas, NumPy

Databases: PostgreSQL, pgvector, Supabase, MySQL

Testing & DevOps: JUnit, pytest, Playwright, Git, GitHub Actions, Docker

Spoken: Brazilian Portuguese (native), English (fluent), Spanish (intermediate, B2)

EXPERIENCE

Research Assistant, Mathematics Potential Lab

Jan 2026 – Present

The University of North Carolina at Chapel Hill

Chapel Hill, NC

- Coded **50+** full-text articles from **6** mathematics and special education journals in **NVivo**, building a qualitative research database on inclusive math instruction
- Analyzed interview feedback from **3** teachers on research-backed educational resources; now designing a second interview round to reach more teachers
- Presented **2** research posters and a **50-minute** talk at Appalachian State University's Inclusion Summit

Tech Intern

Jan 2022 – Jul 2023

FCamara Consulting and Training

São Paulo, SP

- Developed and optimized a **Java**, **Spring Boot**, and **PostgreSQL** employee timekeeping system, improving timesheet data accuracy by **30%** and reducing payroll processing time by **40%** for **2,000+** employees

Research Assistant

Jun 2020 – Dec 2023

Federal Institute of Education, Science, and Technology

Itaporanga, PB

- Developed and deployed a low-cost access-tracking prototype for the campus facilities, reducing manual logging time by **85%** and cutting per-unit hardware cost by **60%**, by integrating **Groovy**, **Ruby on Rails**, and **Arduino**

PROJECTS

Cupinudous | *Java, Spring Boot, Python, FastAPI, CUDA, PostgreSQL, Claude API*

Sep 2026 – Present

- Building a web app that turns research-backed learning techniques into practical study steps for CS and math
- Developing a **RAG** pipeline with the **Claude API** and **pgvector** to ground study steps in cited research
- Writing custom **CUDA** kernels for GPU-accelerated similarity search over embeddings, with a **NumPy** fallback
- Architected **Spring Boot** and **FastAPI** microservices with **Docker Compose** and **GitHub Actions** CI

Fever | *TypeScript, React Native, Expo, Python, Supabase, LightGBM*

Aug 2026 – Present

- Building a cross-platform iOS/Android dating app for UNC students, restricting signup to verified @unc.edu emails via **Supabase Auth**
- Developing an AI matchmaking pipeline with **Claude Haiku 4.5**, **sentence-transformers**, and **LightGBM** to pair users on personality, not looks
- Replacing infinite swiping with weekly match drops using Supabase jobs and **Expo** push notifications

Programming and Robotics, Tocando em Frente NGO | *Arduino, Scratch, Python*

Dec 2021 – Nov 2024

- Founded the program and led a **5-person** core team, bringing tech education to underserved students in Brazil
- Designed and spearheaded programming and robotics educational initiatives, impacting **10,300+** students
- Recruited and trained **60+** volunteer instructors